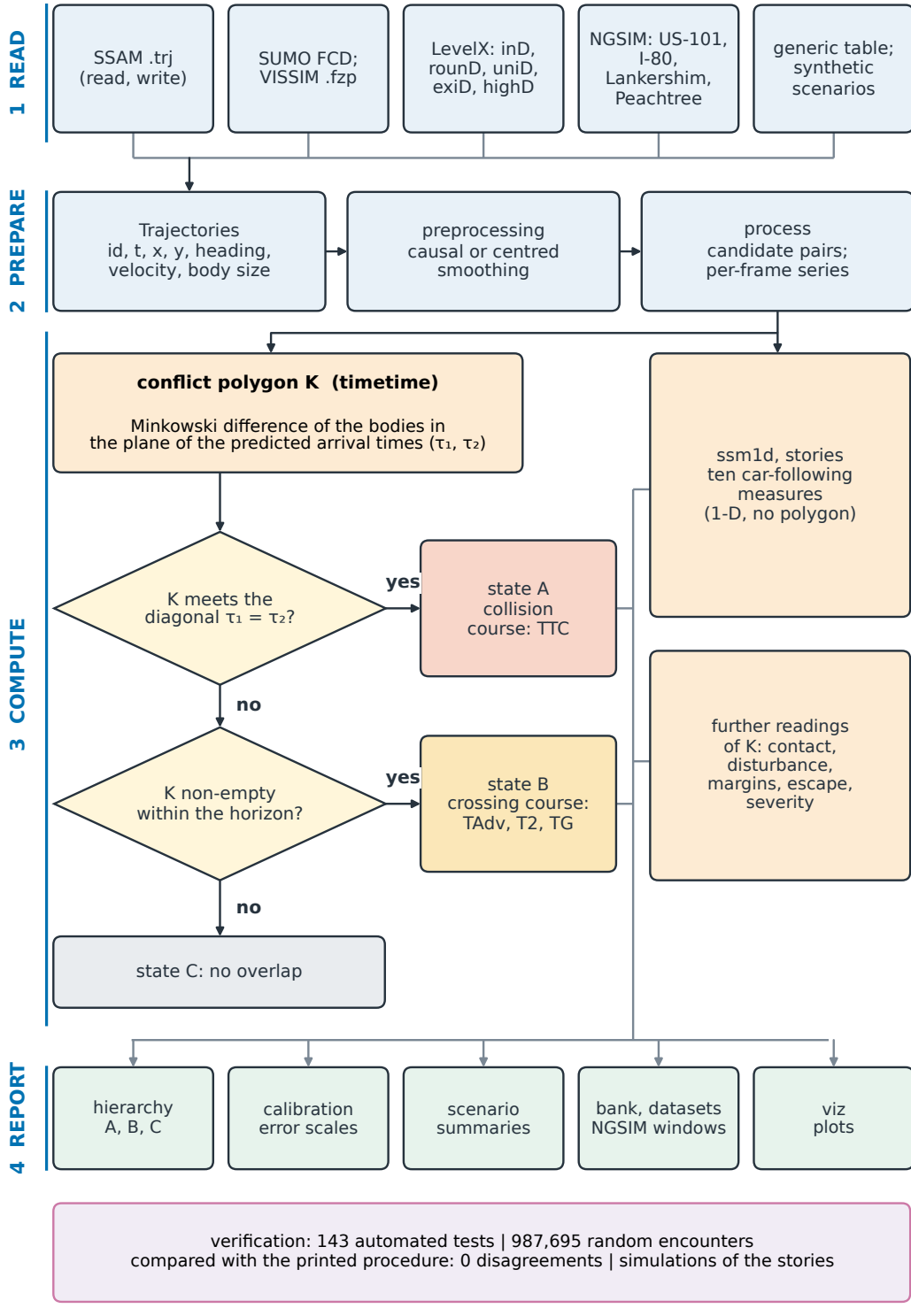
1 READ
SSAM .trj (read, write)
SUMO FCD; VISSIM .fzp
LevelX: inD, rounD, uniD, exiD, highD
NGSIM: US-101, I-80, Lankershim, Peachtree
generic table; synthetic scenarios
2 PREPARE
Trajectories id, t, x, y, heading, velocity, body size
preprocessing causal or centred smoothing
process candidate pairs; per-frame series
3 COMPUTE
conflict polygon K (timetime)
Minkowski difference of the bodies in the plane of the predicted arrival times (τ1, τ2)
K meets the diagonal τ1 = τ2?
yes
state A collision course: TTC
no
K non-empty within the horizon?
yes
state B crossing course: TAdv, T2, TG
no
state C: no overlap
ssm1d, stories ten car-following measures (1-D, no polygon)
further readings of K: contact, disturbance, margins, escape, severity
4 REPORT
hierarchy A, B, C
calibration error scales
scenario summaries
bank, datasets NGSIM windows
viz plots
verification: 143 automated tests | 987,695 random encounters
compared with the printed procedure: 0 disagreements | simulations of the stories